Week 7 – SQL Query Implementation for E-Commerce Database

Organic Grocery Store Management System tables: Customer, Product, Category, Orders, Order_Details, Payment, and Review.

1. Display Data

1. Display all customer details

```
SQL> SELECT * FROM CUSTOMER;

CUSTOMER_ID FIRST_NAME
----------- ----------------------
LAST_NAME
----------------------------------
EMAIL
----------------------------------
----
PHONE
---------------
ADDRESS
----------------------------------
----
          1 Arun
Kumar
arun@gmail.com

CUSTOMER_ID FIRST_NAME
----------- ----------------------
LAST_NAME
----------------------------------
EMAIL
----------------------------------
----
PHONE
---------------
ADDRESS
----------------------------------
----
9999999999
Chennai
```

```
----------------------------------------------
          2 Priya
Raj
priya@gmail.com

CUSTOMER_ID FIRST_NAME
----------- ------------------------------
LAST_NAME
----------------------------------------------
EMAIL
----------------------------------------------
PHONE
---------------
ADDRESS
----------------------------------------------
9876543211
Coimbatore


CUSTOMER_ID FIRST_NAME
----------- ------------------------------
LAST_NAME
----------------------------------------------
EMAIL
----------------------------------------------
PHONE
---------------
ADDRESS
----------------------------------------------
          3 Kavin
M
kavin@gmail.com

CUSTOMER_ID FIRST_NAME
----------- ------------------------------
LAST_NAME
----------------------------------------------
EMAIL
----------------------------------------------
PHONE
---------------
ADDRESS
----------------------------------------------
9876543212
Madurai
```

Display all products

```
SQL> SELECT * FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
---------- ---------- -------------- ----------
       101
Organic Apple
          1        150             30          1

       102
Organic Banana
          1         80             50          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
---------- ---------- -------------- ----------

       103
Organic Carrot
          2         60             40          2

       104
Brown Rice

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
---------- ---------- -------------- ----------
          3        120             25          2

       105
Green Tea
          4        200             20          3
```

3. Display product names and prices

```
SQL> SELECT Product_Name, Price FROM Product;

PRODUCT_NAME
----------------------------------------
----
     PRICE
----------
Organic Apple
       150

Organic Banana
        80

Organic Carrot
        60


PRODUCT_NAME
----------------------------------------
----
     PRICE
----------
Brown Rice
       120

Green Tea
       200
```

4. Display all orders

```
SQL> SELECT * FROM Orders;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----------- --------- ------------ -------------
       101           1 01-SEP-26          500 Delivered
       102           2 02-SEP-26          750 Pending
       103           1 03-SEP-26          300 Shipped
       104           3 04-SEP-26          900 Delivered
       105           2 05-SEP-26          400 Pending
```

5. Retrieve payment details

```
SQL> SELECT * FROM Payment;

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_ST
---------- ---------- --------- -------------------- ----------
TRANSACTION_AMOUNT
------------------
         1        101 01-SEP-26 UPI                  Success
               500

         2        102 02-SEP-26 Card                 Failed
               750

         3        103 03-SEP-26 Cash                 Success
               300


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_ST
---------- ---------- --------- -------------------- ----------
TRANSACTION_AMOUNT
------------------
         4        104 04-SEP-26 Net Banking          Pending
               900

         5        105 05-SEP-26 UPI                  Success
               400
```

2. Apply Filtering Conditions Using WHERE

1. Products with price greater than ₹5,000

```
SQL> SELECT * FROM Product WHERE Price > 5000;

no rows selected
```

2. Products available in stock

```
SQL> SELECT * FROM Product WHERE Stock_Quantity > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
---------- ---------- -------------- ----------
       101
Organic Apple
          1        150             30          1

       102
Organic Banana
          1         80             50          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
---------- ---------- -------------- ----------

       103
Organic Carrot
          2         60             40          2

       104
Brown Rice

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
---------- ---------- -------------- ----------
          3        120             25          2

       105
Green Tea
          4        200             20          3
```

3. Completed orders

```
SQL> SELECT * FROM Orders WHERE Order_Status = 'Delivered'

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----------- --------- ------------ ------------
       101           1 01-SEP-26          500 Delivered
       104           3 04-SEP-26          900 Delivered
```

4. Products with rating above 4

```
SQL> SELECT p.Product_Name, r.Rating FROM Product p JOIN Review r ON p.Produ
ct_ID = r.Product_ID WHERE r.Rating > 4;

PRODUCT_NAME
--------------------------------------------------------------------------
----
    RATING
----------
Organic Apple
         5

Brown Rice
         5
```

3. Sort Data Using ORDER BY

1. Products from lowest to highest price

```
SQL> SELECT * FROM Product ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------
CATEGORY_ID       PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- --------------- ----------
         103
Organic Carrot
           2          60              40          2

         102
Organic Banana
           1          80              50          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------
CATEGORY_ID       PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- --------------- ----------

         104
Brown Rice
           3         120              25          2

         101
Organic Apple

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------
CATEGORY_ID       PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- --------------- ----------
           1         150              30          1

         105
Green Tea
```

2. Customers alphabetically

```
SQL> SELECT * FROM Customer ORDER BY First_Name ASC;

CUSTOMER_ID FIRST_NAME
----------- ----------------------------------------
LAST_NAME
----------------------------------------------------
EMAIL
----------------------------------------------------
PHONE
---------------
ADDRESS
----------------------------------------------------
          1 Arun
Kumar
arun@gmail.com

CUSTOMER_ID FIRST_NAME
----------- ----------------------------------------
LAST_NAME
----------------------------------------------------
EMAIL
----------------------------------------------------
PHONE
---------------
ADDRESS
----------------------------------------------------
9999999999
Chennai


CUSTOMER_ID FIRST_NAME
----------- ----------------------------------------
LAST_NAME
----------------------------------------------------
EMAIL
----------------------------------------------------
PHONE
---------------
ADDRESS
----------------------------------------------------
          3 Kavin
M
kavin@gmail.com
```

```
CUSTOMER_ID FIRST_NAME
----------- ----------------
LAST_NAME
----------------------------
EMAIL
----------------------------
PHONE
---------------
ADDRESS
----------------------------
9876543212
Madurai

CUSTOMER_ID FIRST_NAME
----------- ----------------
LAST_NAME
----------------------------
EMAIL
----------------------------
PHONE
---------------
ADDRESS
----------------------------
          2 Priya
Raj
priya@gmail.com

CUSTOMER_ID FIRST_NAME
----------- ----------------
LAST_NAME
----------------------------
EMAIL
----------------------------
PHONE
---------------
ADDRESS
----------------------------
9876543211
Coimbatore
```

3. Top 3 expensive products

```
SQL> SELECT * FROM (SELECT * FROM Product ORDER BY Price DESC) WHERE ROWNUM <= 3;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- -------------- ----------
       105
Green Tea
         4        200             20          3

       101
Organic Apple
         1        150             30          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- -------------- ----------

       104
Brown Rice
         3        120             25          2
```

4. Latest orders first

```
SQL> SELECT * FROM Orders ORDER BY Order_Date DESC;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----------- --------- ------------ ------------
       105           2 05-SEP-26          400 Pending
       104           3 04-SEP-26          900 Delivered
       103           1 03-SEP-26          300 Shipped
       102           2 02-SEP-26          750 Pending
       101           1 01-SEP-26          500 Delivered
```

4. Retrieve Unique Values Using DISTINCT

1. Unique product categories

```
SQL> SELECT DISTINCT Category_ID FROM Product;

CATEGORY_ID
-----------
          1
          2
          4
          3
```

2. Different payment methods

```
SQL> SELECT DISTINCT Payment_Mode FROM Payment;

PAYMENT_MODE
--------------------
Net Banking
Card
Cash
UPI
```

5. Search Products Based on Conditions

1. Products priced between ₹20,000 and ₹40,000

```
SQL> SELECT * FROM Product WHERE Price BETWEEN 20000 AND 40000;

no rows selected
```

2. Products belonging to a particular category

Example: Category ID 1.

```
SQL> SELECT * FROM Product WHERE Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- -------------- ----------
       101
Organic Apple
          1        150             30          1

       102
Organic Banana
          1         80             50          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- -------------- ----------
```

3. Products available in inventory

```
SQL> SELECT * FROM Product WHERE Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
---------- ---------- -------------- ----------
       101
Organic Apple
         1        150             30          1

       102
Organic Banana
         1         80             50          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
---------- ---------- -------------- ----------


SQL> SELECT p.Product_Name, i.Stock_Quantity FROM Product p JOIN Inventory I ON p.Product_ID = i.Product_ID WHERE i.Stock_Quantity > 0;

PRODUCT_NAME
--------------------------------------------------------------------------------
STOCK_QUANTITY
--------------
Organic Apple
           30

Organic Banana
           50

Organic Carrot
           40
```

4. Search products by name

Example: products containing “Apple”.

```
SQL> SELECT * FROM Product WHERE Product_Name LIKE '%Apple%';

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
---------- ---------- -------------- ----------
       101
Organic Apple
         1        150             30          1
```

5. Find low-stock products

Example: stock below 10.

```
SQL> SELECT * FROM Product WHERE Stock_Quantity < 10;

no rows selected
```

6. Retrieve Customer and Product Information

1. Customer details with their orders

```
SQL> SELECT c.First_Name, c.Last_Name, o.Order_ID, o.Order_Date, o.Total_Amount FRO
M Customer c JOIN Orders o ON c.Customer_ID = o.Customer_ID;

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
Arun
Kumar                                                   101 01-SEP-26
         500

Priya
Raj                                                     102 02-SEP-26
         750

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------

Arun
Kumar                                                   103 03-SEP-26
         300

Kavin
M                                                       104 04-SEP-26

FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
         900

Priya
Raj                                                     105 05-SEP-26
         400
```

2. Product details with category information

```
SQL> SELECT p.Product_Name, p.Price,c.Category_Name FROM Product p JOIN Category c
ON p.Category_ID = c.Category_ID;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_NAME
---------- --------------------------------------------------
Organic Apple
       150 Fruits

Organic Banana
        80 Fruits

Organic Carrot
        60 Vegetables


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_NAME
---------- --------------------------------------------------
Brown Rice
       120 Grains

Green Tea
       200 Beverages
```

3. Find customers who purchased a specific product

```
SQL> SELECT DISTINCT c.First_Name, c.Last_Name FROM Customer c JOIN Orders o ON c.C
ustomer_ID = o.Customer_ID JOIN Order_Details od ON o.Order_ID = od.Order_ID WHERE
od.Product_ID = 101;

no rows selected
```

7. Apply Multiple Filtering Conditions

```
SQL> SELECT p.Product_Name, p.Price FROM Product p JOIN Category c ON p.Category_ID
 = c.Category_ID WHERE c.Category_Name = 'Fruits' AND p.Price > 100;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Organic Apple
       150
```

2. Products containing the word “APPLE”

```
SQL> SELECT * FROM Product WHERE Product_Name LIKE '%Apple%';

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- -------------- ----------
       101
Organic Apple
          1        150             30          1
```

3. Orders placed within a specific date range

```
SQL> SELECT * FROM Orders WHERE Order_Date BETWEEN DATE '2026-09-01' AND DATE '2026
-09-30';

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----------- --------- ------------ --------------------
       101           1 01-SEP-26          500 Delivered
       102           2 02-SEP-26          750 Pending
       103           1 03-SEP-26          300 Shipped
       104           3 04-SEP-26          900 Delivered
       105           2 05-SEP-26          400 Pending
```

8. Generate Basic Business Reports

Report 1: Product Availability Report

Display product name, price, stock quantity, and availability status.

```
SQL> SELECT Product_Name,Price,Stock_Quantity,CASE WHEN Stock_Quantity > 0 THEN 'Av
ailable' ELSE 'Out of Stock' END AS Availability_Status FROM Product;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Organic Apple
       150             30 Available

Organic Banana
        80             50 Available

Organic Carrot
        60             40 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY AVAILABILITY
---------- -------------- ------------
Brown Rice
       120             25 Available

Green Tea
       200             20 Available
```

Report 2: Customer Report

1. Total number of customers

```
SQL> SELECT COUNT(*) AS Total_Customers FROM Customer;

TOTAL_CUSTOMERS
---------------
              3
```

2. New customer registrations

Your Customer table does not have a registration-date column. To count registrations during September 2026, you would need a date column such as Registration_Date.

Report 3: Order Report

1. Total orders

```
SQL> SELECT COUNT(*) AS Total_Orders FROM Orders;

TOTAL_ORDERS
------------
           5
```

2. Completed orders

```
SQL> SELECT COUNT(*) AS Completed_Orders FROM Orders WHERE Order_Status = 'Delivere
d';

COMPLETED_ORDERS
----------------
               2
```

3. Pending orders

```
SQL> SELECT COUNT(*) AS Pending_Orders FROM Orders WHERE Order_Status = 'Pending';

PENDING_ORDERS
--------------
             2
```

4. Cancelled orders

Your current Order_Status constraint does not allow Cancelled. This query will work after you update the constraint and add cancelled order records.

```
SQL> SELECT COUNT(*) AS Cancelled_Orders FROM Orders WHERE Order_Status = 'Cancelle
d';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

1. Highest-priced products

```
SQL> SELECT * FROM Product ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- -------------- ----------
       105
Green Tea
          4        200             20          3

       101
Organic Apple
          1        150             30          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- -------------- ----------

       104
Brown Rice
          3        120             25          2

       102
Organic Banana

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_ID      PRICE STOCK_QUANTITY  SELLER_ID
----------- ---------- -------------- ----------
          1         80             50          1

       103
Organic Carrot
          2         60             40          2
```

2. Most-reviewed products

```
SQL> SELECT p.Product_Name, COUNT(r.Review_ID) AS Total_Reviews FROM Product p JOIN
 Review r ON p.Product_ID = r.Product_ID GROUP BY p.Product_ID, p.Product_Name ORDE
R BY Total_Reviews DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_REVIEWS
-------------
Organic Apple
            2

Organic Banana
            1

Organic Carrot
            1


PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_REVIEWS
-------------
Brown Rice
            1
```

3. Available products

```
SQL> SELECT Product_Name, Price, Stock_Quantity FROM Product WHERE Stock_Quantity >
 0;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Organic Apple
       150             30

Organic Banana
        80             50

Organic Carrot
        60             40


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Brown Rice
       120             25

Green Tea
       200             20
```